ELISA

Overview

Overnight cell culture medium quickly screened for His-tagged protein using the Cayman His-Tag Detection ELISA Kit (Item No. 10012445). The assay shows whether the target protein is present and whether it is gained or lost at each purification step.

A. Materials

a. His ELISA Monoclonal Antibody
b. His-AP Tracer
c. His-Protein ELISA Standard
d. TBS Assay Buffer Concentrate, 10X
e. AP Wash Buffer Concentrate, 150X
f. Goat Anti-Mouse IgG-coated 96-well plate
g. 96-well cover sheet
h. pNPP Substrate Solution
i. Plate reader capable of reading 405-420 nm
j. UltraPure water
k. Orbital shaker
l. Microcentrifuge
m. Clean test tubes

B. Procedures:

1. Preparation:
    - For TBS Buffer dilute 1ml of TBS Assay Buffer Concentrate (10X) with 9 mL UltraPure water and store at 4°C
    - For AP Wash Buffer dilute 0.33ml AP Wash Buffer Concentrate (150X) with 49.66 mL UltraPure water and store at 4°C
    - For His-AP Tracer reconstitute with 6mL TBS Assay Buffer and store at 4°C
    - For His ELISA Monoclonal Antibody reconstituted with 6mL TBS Assay Buffer and stored at 4°C
2. Sample Preparation do a buffer compatibility check and prepare at least 2 dilutions of each sample
3. Standard Preparation
    - Reconstitute standard with 400 µL TBS Assay Buffer and store at 4°C
    - Label six clear tubes #1~#6, add 940 µL TBS Assay Buffer to tube #1 and 500 µL to tubes #2-6, transfer 60 µL of bulk standard to tube #1 and mix thoroughly (3 µg/mL).

- Serially transfer 500 µL from tube #1 → #2 → #3 → #4 → #5 → #6, mixing thoroughly at each step.

Tube concentration

Tube	S1	S2	S3	S4	S5	S6
Concentration (µg/mL)	3	1.5	0.75	0.375	0.187	0.093

4. For Plate Layout each plate set up must contain: 2 Blank (Blk) wells, 2 non-specific binding (NSB) wells, 2 maximum binding (B_0) wells, a six-point standard curve (S1-S6) in duplicate
5. Assay each sample at two dilutions, each in duplicate and return unused strips to sealed packet and store at 4°C

Reagent addition

Well	TBS Assay Buffer	Standard/Sample	AP Tracer	Anti-His Antibody
Blk	-	-	-	-
NSB	100 µL	-	50 µL	-
B_0	50 µL	-	50 µL	50 µL
Standard/Sample	-	50 µL	50 µL	50 µL

6. Cover the plate with plastic cover sheet and incubate for 90 minutes at room temperature on an orbital shaker
7. Empty wells and rinse 4 times with Wash Buffer, after last wash, gently tap the inverted plate on absorbent paper to remove residual buffer
8. Add 200 µL pNPP Substrate Solution to each well including the Blk and cover the plate and incubate at room temperature for 60 minutes
9. Wipe the bottom of the plate with clean tissue and read at 405-420 nm
10. Check absorbance periodically until B_0 wells reach 0.3 A.U. and read the plate when B_0 wells fall within 0.3-1.5 A.U.
11. Calculate data by
    a. Subtract Blk absorbance from all wells including NSB
    b. Average the NSB wells
    c. Average the B_0 wells
    d. Corrected B_0 = average B_0 - average NSB
    e. For each standard and sample well, calculate B/B_0
    f. Multiply by 100 for $\%B/B_0$

12. To get the Standard curve plot $\%B/B_0$ against His-protein concentration for S1-S6 and apply a four-parameter logistic fit
13. Calculate Sample concentration by
    a. Calculate B/B_0 (or $\%B/B_0$) for each sample and read the concentration from the standard curve equation
    b. Multiply by the dilution factor applied before adding the sample to the well
    c. Re-assay samples with $\%B/B_0$ >80% or <20%, as they fall outside the linear range.
    d. A ≥20% disparity between the apparent concentrations of two dilutions indicates assay interference.
    e. Purify the sample before re-analysis.